Make the situation clear before calculating

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Reference point / ES: punto de referencia / PT: ponto de referência

The zero everything is measured from. Mark it on your sketch.

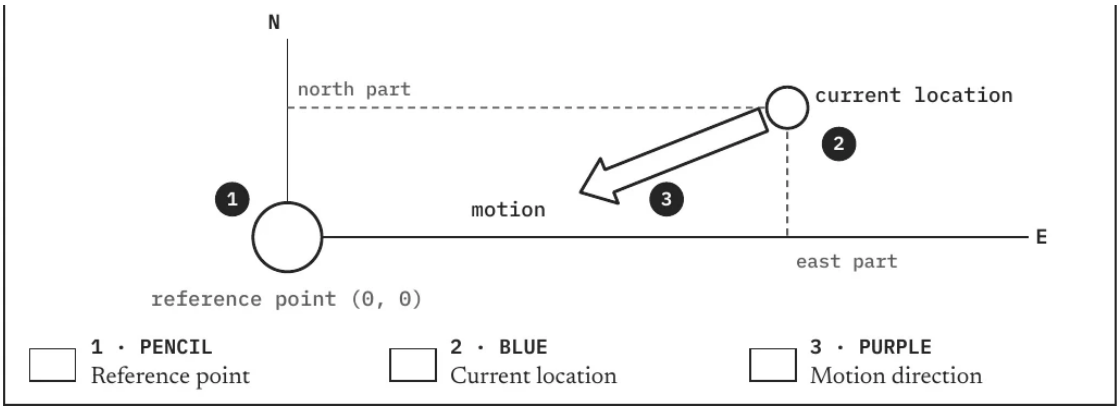
Also on your note page: Known quantity, Model assumption, Range.

B READ THE RELATIONSHIP

A reference frame tells us where zero is and which directions count as positive.

Define the frame, convert units, then forecast on each axis

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Check a prediction: an object starts at 2 m and moves at a steady +3 m/s for 4 s.

Given $x_i = 2$ m $v = +3$ m/s $t = 4$ s · steady v

Rule $x_f = x_i + v \times t$

Work change $= 3 \times 4 = +12$ m $x_f = 2 + 12$

Answer 14 m check: direction + size + start

E SAVE FOR THE FINAL MEETING

Now it's your forecast. Pick the evidence on XJ that matters, choose a model, and plan the calculation that answers the unit question.

Read the physics: Reference point

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Name a reference point, mark the origin and choose axes. Record the starting position and the time. Include a unit with each value.

02 HOW IT WORKS

If velocity is steady, add velocity times elapsed time to the starting coordinate. In two dimensions, do that once for x and once for y. Use the matching component each time.

03 WHAT TO CHECK

Check whether the final position is on the expected side of zero. Say which motion assumption could fail when a new sighting arrives.

04 TRY THE EXAMPLE FROM YOUR NOTES

Check a prediction: an object starts at 2 m and moves at a steady +3 m/s for 4 s. 14 m check: direction + size + start.

05 CONNECT TO XJ

Now it's your forecast. Pick the evidence on XJ that matters, choose a model, and plan the calculation that answers the unit question.

SAY IT BEFORE YOU WRITE

Would moving zero change the motion or only its coordinates? From ___ as zero, I predict ___ at time ___. I assumed ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Reference point

ES: punto de referencia / PT: ponto de referência

Meaning in my words	How can you tell?
How would you explain reference point to a friend? In my words, this means ___.	A sign says “5 m east.” What place do we need to identify? I can tell because ___.
Example or sketch	**Non-example and why**
How would you mark a cart 3 m west of a doorway chosen as zero? My example shows ___.	If I choose the table as zero instead, did the cart actually move? I would change ___ because ___.

Known quantity

ES: magnitud conocida / PT: grandeza conhecida

Meaning in my words	How can you tell?
What makes a number a known quantity in a problem? In my words, this means ___.	A cart starts at rest and is timed for 3 s. Which two values can you write down? I can tell because ___.
Example or sketch	**Non-example and why**
A sensor reads +4 m at the start. How would you enter that known value? My example shows ___.	A friend lists a guessed final speed among the given values. Where should it go? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Model assumption

ES: supuesto del modelo / PT: hipótese do modelo

Meaning in my words	How can you tell?
How would you explain model assumption to a friend? In my words, this means ___.	What must be true to predict another 2 m each second from a cart's earlier motion? I can tell because ___.
Example or sketch	**Non-example and why**
Write an assumption beside a straight-line forecast for XJ. My example shows ___.	Is tomorrow's camera reading an assumption once we measure it? I would change ___ because ___.

Range

ES: intervalo / PT: intervalo

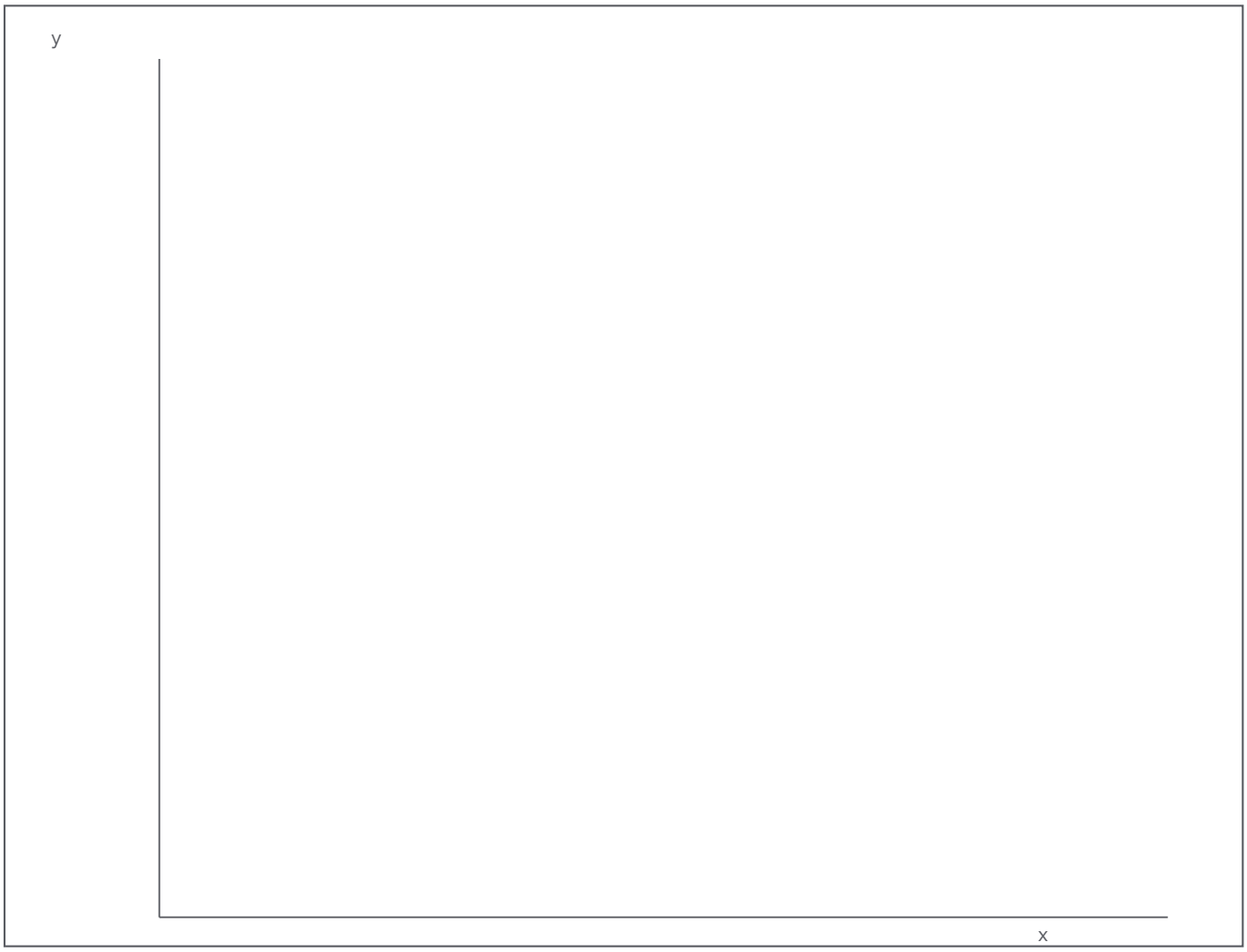
Meaning in my words	How can you tell?
How would you explain range to a friend? In my words, this means ___.	What positions are included in 12 ± 2 m? I can tell because ___.
Example or sketch	**Non-example and why**
How would you mark the forecast on a number line? My example shows ___.	A partner says only 12 m counts as a match. What do the endpoints show? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch the reference frame and one sideways component. Box your time unit and one model assumption.

Labels you can use: origin · direction · assumption.



Tell a partner

From ___ as zero, I predict ___ at time ___. I assumed ___.

Two reports. One sighting.

21

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare position reports that use different zeros.

CHOOSE ONE JOB TOGETHER

SURVEY ENGINEER ☐

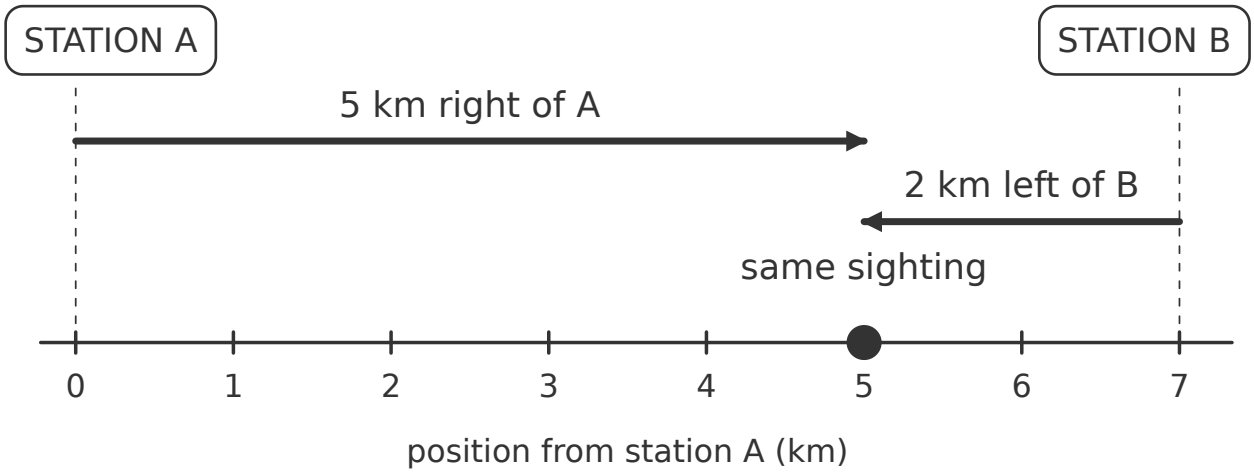
Use one axis for both stations. Give dispatch the position from A, then name what a future forecast still needs.

MAP BROADCASTER ☐

Show both stations and zeros on the locator map. Explain whether the two reports describe different locations.

Word help: zero / frame / position

Two observatories file the same sighting: "**5 km** right of station A" and "**2 km** left of station B." B is **7 km** right of A.



01 OUR FIRST CALL

Do the reports agree? Give one location and explain how the map supports it.

From station ___, the position is ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What number changes in the new frame? What physical location stays the same?

Now we would say ___ because ___.

Switch speakers. Are the two stations using the same zero?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks